

TitanZero — Titan Pulse Automation Engine

Purpose

Titan Pulse is the automation and orchestration engine of the TitanZero platform.

It listens to signals emitted by Titan Signal and executes automation rules in response. Pulse allows the system to perform routine operational actions automatically without human intervention.

Titan Pulse enables:

workflow automation

operational orchestration

lifecycle progression

reactive business processes

AI-assisted execution

All Pulse actions remain fully auditable and may be escalated to Titan Zero governance when risk thresholds are exceeded.

Pulse Philosophy

Pulse follows a simple operational principle:

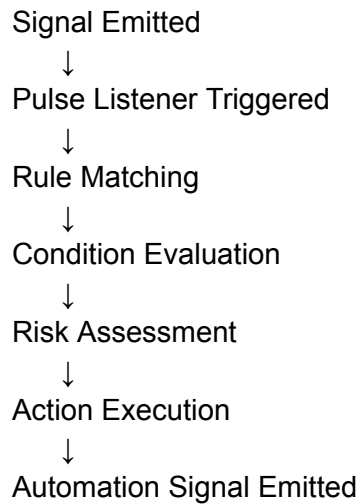
> When a signal occurs, evaluate rules. If conditions match, execute actions.

Automation therefore becomes a deterministic response to system signals.

Pulse does not invent actions. It executes defined rules safely and predictably.

Automation Flow

Automation operates in the following sequence:



This ensures that automation remains transparent and traceable.

Pulse Engine Location

In the application structure:

app/Core/TitanPulse

Core components include:

PulseEngine
PulseRuleEvaluator
PulseConditionResolver
PulseActionExecutor
PulseRiskEvaluator

Each component performs a specific role in the automation pipeline.

Rule Structure

Automation rules are composed of three parts.

Trigger
Condition
Action

Example rule:

Trigger: job.completed
Condition: job.recurring = true
Action: schedule_next_visit

Rule Storage

Rules are stored in the database.

tz_automation_rules

Example structure:

id
team_id
trigger_event
conditions
actions
priority
enabled
created_at

Each rule belongs to a tenant and applies only within that tenant's environment.

Rule Execution Tracking

All rule executions are recorded.

tz_automation_runs

Fields:

id
team_id
rule_id
signal_id
execution_status
result
created_at

This table ensures complete auditability of automation actions.

Condition Evaluation

Conditions determine whether a rule should execute.

Conditions may reference:

entity attributes

signal payload data

tenant configuration

time conditions

Example:

job.recurring = true
invoice.amount > 500
customer.status = VIP

Pulse resolves conditions using the PulseConditionResolver.

Action Execution

Actions represent the tasks performed by Pulse.

Actions may include:

create_record
update_record
send_message
schedule_job
assign_user
emit_signal
trigger_campaign

Example action:

```
send_message(customer, "Your service has been completed")
```

Actions are executed through the PulseActionExecutor.

Automation Graph

Pulse rules form a network of automation flows.

Example:

```
job.completed  
↓  
emit signal  
↓  
invoice.create  
↓  
invoice.sent  
↓  
payment.reminder automation
```

This creates chained workflows across the system.

Safety and Risk Evaluation

Before executing an action, Pulse evaluates the risk level.

Risk levels:

LOW

MEDIUM

HIGH

Low-risk actions execute automatically.

Medium-risk actions may require confirmation.

High-risk actions trigger Titan Zero governance review.

Titan Zero Escalation

When Pulse encounters high-risk actions:

automation.high_risk

A signal is emitted.

Titan Zero then activates Trinary Bayesian Mode to evaluate the action.

Possible outcomes:

approve

modify

reject

Pulse executes the approved action.

Idempotency

Automation rules must be idempotent.

This means executing the same rule multiple times should not create duplicate or inconsistent results.

Pulse ensures this by checking:

rule execution history

entity state

signal timestamps

Retry Logic

Failed automation runs may be retried.

Retry logic includes:

retry_count

retry_delay

failure_reason

This allows temporary issues (such as API outages) to recover automatically.

Manual Overrides

Operators may override automation through Titan Command.

Overrides generate governance signals:

automation.override

These signals are monitored by Titan Zero.

Automation Monitoring

Pulse provides operational visibility through monitoring dashboards.

Operators can see:

active automation rules

rule execution history

failure alerts

high-risk escalations

These insights help maintain confidence in automated operations.

Integration with Lifecycle Assistants

Lifecycle assistants interact with Pulse by:

generating rule suggestions

triggering automation actions

responding to automation outcomes

Example:

job.completed



Pulse triggers invoice



Lifecycle assistant sends follow-up message

This allows automation and AI reasoning to cooperate.

Integration with Titan Signal

Pulse listens to signals and emits new signals.

Example:

job.completed
↓
Pulse executes rule
↓
invoice.created
↓
signal emitted

This ensures that the entire system remains signal-driven.

Summary

Titan Pulse is the automation engine of TitanZero.

It listens to system signals, evaluates automation rules, executes actions, and records all activity for auditing and governance.

Pulse enables TitanZero to operate as a self-orchestrating business platform, while Titan Zero governance ensures that higher-risk decisions remain safe and controlled.